

**STRUCTURAL GROUP CHAIRMAN ADDENDUM TO
FACTUAL REPORT**
(14 pages)

1. IDENTIFICATION OF OVERSIGHT ISSUES DIRECTLY RELATED TO THE ACCIDENT

1.1 CEILING DESIGN

The ceiling design consisted of concrete ceiling panels that measured 12 feet wide by 8 feet long by 4 inches thick. A ceiling module was referred to as five (5) concrete ceiling panels that were located side by side and oriented parallel to the travel lanes. The total length of a ceiling module was 40 feet.

The structural members used to support the concrete ceiling panels were panel supports, hanger supports, and adhesive anchors embedded into the tunnel ceiling. The vertical hanger supports and adhesive anchors worked in pure tension. The angled (45 degree) hanger supports prevented movement of the ceiling system in the longitudinal direction (parallel to the travel lanes) while the struts prevented movement of the ceiling system in the transverse direction.

The predominant hanger support was the vertical hanger that contained 2 adhesive anchors. The vertical hanger supports with an angled (45 degree) hanger support on either side contained 4 adhesive anchors. At least one vertical hanger support with an angled (45 degree) hanger support was located along each of the panel supports (one along the north and south panel supports and two along the middle – north end and middle – south end panel supports). The function of the angled (45 degree) hanger supports, was not only to prevent movement of the ceiling system in the longitudinal direction, but also to transfer a significant amount of load from the adjacent 2 anchor vertical hangers to the 4 anchor vertical hanger. This is one explanation of why the 4 anchor vertical hanger took a significant higher load than the adjacent 2 anchor vertical hangers.

The ceiling design did provide some measure of redundancy. In the event of a loss of support at a hanger location, loads could redistribute longitudinally, through bending in the angled (45 degree) hanger supports. In addition, the presence of 2 adhesive anchors and 4 adhesive anchors at each hanger greatly reduced the probability of a total loss of support in a hanger in the event of an epoxy bond failure or shear failure of a single adhesive anchor.

The use of the heavy ceiling panels (50 pounds per square foot) for the I-90 connector tunnel finishes (C09B2) contract was an outgrowth of a Value Engineering Report¹ to capture the lessons learned from the Ted Williams tunnel finishes (C05B1) contract. The Value Engineering Report indicated the ceiling panels in the Ted Williams tunnel were time consuming and labor intensive to install.

One of the key recommendations from the Value Engineering Report was to study the costs associated with various ceiling options compared to the current design in the Ted Williams tunnel. The various ceiling options were the following:

¹Commonwealth of Massachusetts Central Artery (I-93) / Tunnel (I-90) Project Value Engineering Report – Tunnel Finishes, March 1995, prepared for the Department of Public Works and Bechtel / Parsons Brinckerhoff by Olympic Associates Company.

Option 0	Panel weight 30 lb/sf, cost \$50/sf approximate: porcelain enameled steel face precast panels on a 4' x 6' grid of hangers (current design)
Option I	Panel weight 35 lb/sf, cost \$44.69/sf: precast panels supported on a 4' x 6' grid (current design modified to remove porcelain enamel facing)
Option II	Panel weight 50 lb/sf, cost \$34.55/sf: precast lay-in panels supported on continuous inverted steel tees at 12' on center (tees are aligned with the lane edges)
Option III	Panel weight 50 lb/sf, cost \$46.23/sf: cast-in-place concrete supported on continuous inverted steel tees at 12' on center (tees are aligned with the lane edges)

The Value Engineering Report recommended the following option be selected²:

“B/PB recommends the selection of ceiling system Option II on the basis of maximum cost containment, for an estimated total projectwide savings of \$65 million.”

1.2 USE OF ADHESIVE ANCHORS OVER UNDERCUT ANCHORS

Gannett Fleming, Inc. wrote an internal memorandum of telephone conversation between Bechtel / Parsons Brinckerhoff (B/PB) and Gannett Fleming, Inc. dated January 13, 1997 regarding the use of undercut anchors in the I-90 connector tunnel finishes (C09B2) contract.

B/PB: *In a discussion with Mark Cunningham of Hilti, I was informed that you are considering using undercut anchors in the C09B2 Project.*

GF: *Yes, undercut anchors are, from a design standpoint, a good choice for drill in anchorage in the tension zone of reinforced concrete.*

B/PB: *The problem is proper installation. If not fully undercut with special drills and accurate placement, they can be pulled out by hand. B/PB has already directed Deleuw – Cather to remove them from their contract documents. This directive is CA/T project wide.*

²Commonwealth of Massachusetts Central Artery (I-93) / Tunnel (I-90) Project Value Engineering Report – Tunnel Finishes, March 1995, prepared for the Department of Public Works and Bechtel / Parsons Brinckerhoff by Olympic Associates Company, 2.9 Tunnel Ceiling Alternatives, Summary and Recommendations, sheet 2 of 18.

GF: *C09B2 will re-direct our design efforts to other types of anchorage for the 04A2 portion of C09B2.”*

Gannett Fleming, Inc. sent a letter to Bechtel / Parsons Brinckerhoff (B/PB) dated August 25, 1997. The letter indicated that Gannett Fleming, Inc. had researched various types of post-installed anchoring systems for hanging the ceiling in the C04A2 portion of the C09B2 tunnel finishes contract. The various types of post-installed anchorage systems included the resin bonded anchor, the undercut anchor, and the expansion anchor. The letter mentioned the undercut anchor appeared to have the desired characteristics for supporting the ceiling system in the subject tunnel. The letter indicated the following:

“UNDERCUT ANCHORS

Undercut anchors consist of a threaded rod with nut assembly, expansion sleeve and expansion plug. When the nut is tightened, the expansion plug expands the steel expansion sleeve into the undercut at the top of the drilled hole. Tensile forces are transferred into the concrete by keying and bearing against the undercut in the concrete.

A research report which we reviewed stated that an undercut anchor in cracked concrete will retain 65% to 70% of the anchors ultimate capacity in uncracked concrete. Undercut anchors perform the same as cast-in headed anchors according to this paper. These anchors will perform similar to steel when exposed to high temperatures.

Undercut anchors are recommended by manufacturers for application in the tensile zone of concrete as well as cracked concrete. They are recommended for dynamic loading as well. These anchors are available in a variety of steel materials and with zinc plating or hot-dipped galvanized coatings depending on the manufacturer. These anchors appear to have the desired characteristics for supporting the ceiling system in the subject tunnel.”

NTSB investigators questioned personnel from both Bechtel / Parsons Brinckerhoff (B/PB) and Gannett Fleming, Inc. regarding whether any written correspondence was received or sent subsequent to the August 25, 1997. Both firms could not produce any documentation subsequent to the August 25, 1997 letter indicating approval of the adhesive anchors. It was the opinion of Gannett Fleming, Inc. that Bechtel / Parsons Brinckerhoff (B/PB) directed them to use the adhesive anchors, even though the written correspondence suggests Gannett Fleming, Inc. recommended using the undercut anchors.

Bechtel / Parsons Brinckerhoff (B/PB) knew of problems encountered during installation of the adhesive anchors in the Ted Williams tunnel but still recommended using them in the I-90 connector tunnel finishes (C09B2) contract. The problems encountered included insufficient mixing of the epoxy, considerable moisture in the holes, voids existing within the concrete of the ceiling, overdrilling of the holes, and installation of the epoxy during cold weather.

Powers Fasteners produced a Specification and Design Manual³ that discussed the advantages for using an adhesive anchor and undercut anchor. The advantages for using an adhesive anchor were the following:

“Adhesive anchoring systems offer many advantages for applications requiring high load capacities.....

Chemical or adhesive anchors typically use an ester based resin or an epoxy to bond threaded bolts or reinforcing bars into the anchor hole. Normally, the hole size is only slightly greater than the rod bolt or bar diameter. Typical two-part systems include an encapsulated glass design such as the Chem-Stud or Hammer-Capsule and plastic cartridge injection systems in various configurations such as the Power-Fast+ or AC100 Plus systems.

Adhesive anchors achieve their load capacities based on the ability of the adhesive used to bond to the base material. This ability to bond, often referred to as “wetting action”, will vary depending upon the adhesive type and formulation. While the adhesive bonds to the base material, it also forms a mechanical interlock around the threaded rod. In order to form this interlock, it is important to use anchor rods which have some type of deformation. Examples would be threaded rod or deformed reinforcing bars. Smooth dowel bars coated with epoxy can be used with some types of adhesive systems.”

The advantages for using an undercut anchor were the following:

“Mechanical expansion anchors can usually be loaded immediately after installation which may be an advantage in some applications as compared to bonded adhesive anchors which must be allowed to cure prior to loading. Steel expansion anchors generally have a greater resistance to the effects of elevated temperature when compared with adhesives such as ester based resins or epoxies.....

Undercut anchors expand at the bottom of the drilled hole similar to a compression type anchor except that the actual diameter of the expanded area is wider than the drilled hole, undercutting the base material similar to a dove tail slot. Anchors of this type can be self undercutting or may require a secondary drilling operation to form the undercut at the bottom of the drilled hole. During installation, as the expansion mechanism undercuts the base material, it forms a large bearing area which can transfer greater load to the base material.”

³Architectural and Engineering Specification and Design Manual 5th Edition, Anchoring & Fastening Systems, Powers Fasteners, pages 26, 27, and 183.

1.3 PROOF TEST LOADS FOR THE ADHESIVE ANCHORS

Gannett Fleming, Inc. indicated the source of the proof test load was based on the anchor testing procedure specified for the Ted Williams Tunnel that was furnished in the form of a sample specification by Bechtel / Parsons Brinckerhoff (B/PB) to Gannett Fleming, Inc. via facsimile dated September 23, 1998. The facsimile was used as the basis for the proof test load contained in the special provisions of the I-90 Seaport Access Tunnel Finishes construction contract. The facsimile indicated the following:

“Subject: Anchor Bolt Testing

Comments: Please incorporate/approve these changes based upon lessons learned in TWT.

3.07 INSTALLATION OF ADHESIVE TYPE CEILING ANCHORAGE SYSTEM

F. After chemical adhesive has cured, proof test all anchors to 125 percent of their design load per ASTM E 448, unconfined static tests for tension and shear. Record results of proof tests and submit to Engineer. Clearly mark and abandon anchors which fail test. Proof test procedure must be submitted to and reviewed by Engineer prior to installation.”

The proof test load mentioned in the special provisions of the I-90 Seaport Access Tunnel Finishes⁴ construction contract indicated the following:

“Anchor shall be tension loaded to a service design load of 3.25 kips tension.”

Gannett Fleming, Inc. confirmed that the proof test load of 3.25 kips tension was a number derived from taking 125% of the design service load per anchor (i.e. 1.25 x 2.6 kips) for the two anchor hanger system. The two anchor hanger system was the governing hanger connection.

The proof test load for the adhesive anchors in the Ted Williams Tunnel⁵ was the following:

“Each anchor was tested to 1.25 times its rated maximum working load, equal to 7,000 pounds for 3/4 inch diameter anchors.”

The proof test load of 7,000 pounds (or 7 kips) was a number derived from taking 125% of the maximum working load per anchor (i.e. 1.25 x 5.6 kips) for the two anchor hanger system.

⁴Precast Concrete Ceiling System Section 723.480, I-90 Seaport Access Tunnel Finishes C09B2, April 22, 1998, Div. II Special Provisions, 2.08 (B1) Adhesive Type Ceiling Anchorage System, page 8 of 13.

⁵Metropolitan Highway System Stem to Stern Safety Review – Phase I, Volume 3 – Tunnels – Finishes, November 15, 2006, page 16.

Bechtel / Parsons Brinckerhoff (B/PB) field engineers were on scene continuously during the installation and proof testing of the adhesive anchors and completed Field Engineers Daily Reports (FEDR).

1.4 FIELD ENGINEERS DAILY REPORTS (FEDR) OF PROBLEMS WITH ADHESIVE ANCHORS IN I-90 CONNECTOR TUNNEL

Listed below are excerpts from the Field Engineers Daily Reports (FEDR) that document the problems with the installation and proof testing of the adhesive anchors in the I-90 connector tunnel.

“October 13, 1999

FArch met with J. Martin (MCC) and Jack Armour (Powers Fasteners, Inc.) atop the mock-up ceiling to discuss the problem of anchor bolts “creeping” out over time. We showed Armour the problem (displaced hanger plates/bolts) and reviewed the anchor bolt installation and testing procedures, and module erection process. We gave Armour a copy of FArch’s 10/08/99 field sketch. Armour did not offer any analysis or conclusions at this time, preferring to confer with his company’s engineering staff before offering opinions and/or solutions.

October 21, 1999

At approximately 10:15 a.m. a field meeting was held with Stephen A. Rice, the Field Engineering Manager for Powers Fasteners, from New Rochelle, NY along with the following additional personnel

Neil Joyce – MCC Senior Project Engineer

Jim Martin – MCC QC Manager

Bill Britto – MCC Superintendent

Scott Abbotts – MCC Asst. Supt.

Phil Aikele – B/PB LFE

Jim Barrett – B/PB Field Architect

The subject of the meeting was to discuss and determine the cause of the pull out of 5 of the ceiling anchors in the mock-up at ceiling module HOV078. It should be noted that Powers Fasteners is the manufacturer of the epoxy used to install the 5/8” stainless steel ceiling anchors. Steve reviewed with the MCC personnel the installation and testing procedures that we used for installing the ceiling anchors. Several potential areas of concern were discussed:

- 1) Installation temperature. It was determined that this was not an issue as the epoxy was installed during the months of July & August when the temperatures in the tunnel were considerably above the minimum required.*

- 2) *Reduction in load capacity due to lack of adhesion with cut reinforcing steel. It was discussed that the worse case would be if two #11 bars were penetrated at their intersection. This would reduce the design hole depth to an effective embedment length of 2.75 inches, which is a 50% reduction of the contact area with concrete. According to the manufacturer's data the capacity of the anchor would then be just under 10.0K rather than the 31.0+K if the full effective depth of 5.5 inches were achieved. However, this approximately 10.0K is still greater than the 3.25K at which each anchor was pull tested.*
- 3) *Lack of adhesion due to using a diamond tipped core drilling bit rather than a carbide bit. It was noted that a carbide bit would give a rougher surface inside the hole and hence greater adhesion, but that it was not possible to use a carbide bit due to the amount reinforcing steel.*

November 12, 1999

Two technicians from Conam Inspection Inc. conducted pull out tests on 2 bolts in WB which were previously skipped, and both bolts passed.

The technicians then went to the mock-up area to pull out one of the adhesive anchor bolts which had previously passed the pull-out test, but had been slipping out over the last several months. The bolt had passed the pull-out test on 7/30/99. However, a survey conducted on 10/08/99 found that the bolt had slipped out 9/16", the most displacement of the 5 bolts that were slipping. When the standard pull-out test load (3.25 kips) was applied today, the bolt held for a few seconds, then began to pull out with almost no resistance. The failed bolt was removed and a new bolt will need to be installed and re-tested.

Upon examining the pulled out bolt, it appeared to lack sufficient epoxy to fully fill the drilled hole. Also, there was a significant amount of concrete dust adhered to the epoxy surrounding the bolt, usually an indication that the drilled hole was not completely cleaned out prior to installation.

The epoxy nearest to the embedded tip of the bolt was brittle and easily crumbled, usually an indication of improper mixing of resin and hardener. The epoxy near the exposed end of the bolt (but still in the embedded portion) appeared to be fully and properly cured. This would indicate the mixture of resin and hardener was not uniform during the injection of epoxy into the drilled hole.

In spite of this combination of installation deficiencies it appears the bolt was able to develop just enough strength to pass the pull-out test. However after several weeks of constant loading by the ceiling module, the bond was broken, and the bolt began to slip out.

March 13, 2000

Two technicians from Conam testing labs were testing adhesive anchor bolts in the 4A2 area of HOV, east of the mock-up area. By COB they had tested 72 bolts and there were two failures. The failed bolts held the test load of 6350 lbs when it was first applied. The testing procedure requires the load to be held for two minutes. During the two minute holding period both bolts ‘popped’ loose. Both bolts came out almost perfectly clean, with almost no sign of cured epoxy clinging to either one. At the location of the failed bolts a new bolt will be installed and retested. This completes testing in the HOV 4A2 area, except for the bolts which are currently supporting the Mock-up ceiling panels, which still need to be re-tested to the higher (6350 lb) load.”

1.5 ACTIONS FOLLOWING THE IDENTIFICATION OF PROBLEMS WITH ADHESIVE ANCHORS IN I-90 CONNECTOR TUNNEL

Modern Continental Company (MCC) sent a letter to Bechtel / Parsons Brinckerhoff (B/PB) dated October 7, 1999 indicating a small percentage of the adhesive anchors in the HOV ceiling mockup area appeared to show signs of tensile movement, thus creating gaps in the ceiling hanger plate to ceiling interface. A series of 15 correspondence documents ensued between MCC and B/PB from October 1999 through March 2001 that included letters, contractor transmittals, and repair procedures.

The actions following the identification of problems with adhesive anchors in the I-90 connector tunnel were brought to the attention of the following personnel at B/PB and MTA:

B/PB - Field Architect
B/PB – Lead Field Engineer
B/PB – Resident Engineer
B/PB⁶ – Area Engineer
MTA⁷ – Director of Construction
B/PB⁷ – Deputy Program Manager
B/PB⁷ – Program Manager

⁶The B/PB Area Engineer was carbon copied (cc:) on the 15 correspondence documents between MCC and B/PB from October 1999 through March 2001 that included letters, contractor transmittals, and repair procedures.

⁷A partial email chain dated December 2, 1999 indicated the senior project managers at MTA and B/PB apparently were informed of problems with the ceiling anchors in the C09B2 contract.

Figure 1 shows the project organization chart for the Central Artery / Tunnel Project.

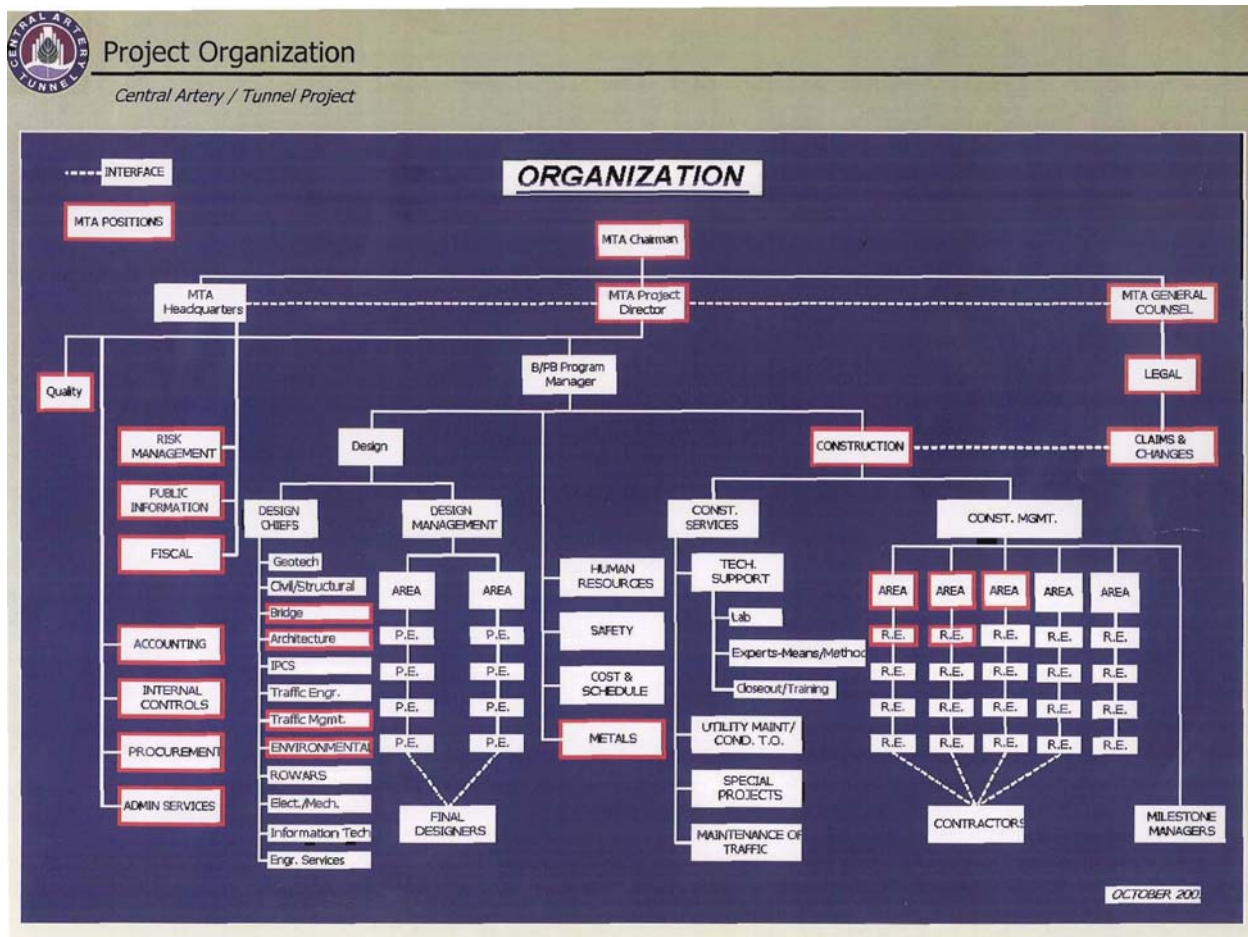


Figure 1 – Project Organization Chart for the Central Artery / Tunnel Project

B/PB personnel attempted to resolve the problems with the adhesive anchors in the I-90 connector tunnel. On December 1, 1999, B/PB personnel issued a Deficiency Report (DR1) to MCC that indicated the following:

“At least five ceiling adhesive anchors in the area of ceiling module HOV 079 have failed, that is pull-out has been observed, since the hanging of the ceiling module. See the attached sketch for the locations of the failed anchors and the amount of pull-out that has been observed.”

The result of the 15 correspondence documents between MCC and B/PB revealed the adhesive anchors in the HOV ceiling mockup area would be retested at a higher load of 6,350 pounds. The higher proof test load of 6,350 pounds was expanded to include the entire HOV lane in the C04A2 construction contract, however, it did not include the eastbound and westbound lanes of the I-90 connector tunnel. There were 187 adhesive anchors retested at the higher load in the HOV lane, of which 19 anchors failed. Each of the failed anchors was

replaced and retested. Of these only one failed upon retesting and it was then replaced and retested again and passed. All of the adhesive anchors in the HOV lane were tested and replaced as necessary and retested until passing at the higher load.

MCC and B/PB disagreed on the proof test value that should be used to test the adhesive anchors in the HOV ceiling mockup area. MCC argued the proof test value should be 3,250 pounds (the value specified in the contract documents) and B/PB argued the proof test value should be 6,350 pounds (the allowable load capacity for the epoxy). MCC requested the disagreement be handled in accordance with the project's issue resolution model and a Senior Partners meeting was scheduled for February 10, 2000 to discuss a resolution plan. To avoid bringing an issue of relatively low cost (\$16,396.50) before the Senior Partners, MCC and B/PB reached an agreement under which all of the failed anchors would be replaced, in accordance with the approved procedures, and proof tested to 6,350 pounds. B/PB agreed to pay the testing cost for any anchor that passed the test to the higher load of 6,350 pounds. B/PB and MCC would split the cost of any anchor that had previously passed the test at 3,250 pounds but failed at 6,350 pounds. MCC was subsequently paid \$16,396.50.

The corrective actions taken by MCC were verified by B/PB in late 2000, and Deficiency Report (DR1) was closed on January 26, 2001.

1.6 EPOXY BEING USED WHEN NOT ON STATE'S QUALIFIED LIST

Specification section 960.050, 2.01.W. Addendum No. 9 covered the material and manufacturer requirements for all anchors and inserts except the ceiling system hanger support adhesive anchors. Specification section 960.050, 2.01.W. Addendum No. 9 indicated the following:

“W. Adhesive (Chemical) Anchors:

- 1. Adhesive: two component systems, such as vinyl urethane methacrylate resin with dibenzoyl peroxide hardener. Adhesive will meet the requirements of Spec. 723.480, 2.08.B.*
- 2. Anchor (threaded rod or dowel) material: Type ASTMA36, carbon steel.*
- 3. Manufacturer and Product Identification:*
 - a. Manufacturer must be pre-approved by MHD per requirements of Section 6.01 in Division I.*
 - b. Hilti Fastening Systems, Inc. (“HVA” Adhesive Anchor System), or equal by Liebig International, Inc. or Rawlplug Co. Inc., or other approved source.”*

Specification section 960.050, 1.05.C. Addendum No. 9 was clear that the specification did not pertain to the ceiling system hanger support adhesive anchors and to see Specification section 723.480 Precast Concrete Ceiling System:

“C. Anchor and Insert Qualification Testing – For all anchors and inserts except ceiling system hanger supports adhesive anchors. (See Proj. Spec. No. 723.480 Precast Concrete Ceiling System).”

Specification section 723.480, 2.08.B. Addendum No. 11 covered the adhesive type ceiling anchorage system. The specification only mentioned performance standards and did not give any recommendation for types of manufacturers as previously mentioned in Specification section 960.050.

“B. Provide chemical adhesive type anchor system to anchor support system to concrete structure. Provide rods, washers and nuts fabricated of compatible steel in the diameters and embedment lengths shown on the drawings. Provide adhesive consisting of 2 component (plastic resin and catalyst hardener) mixture. Resin material shall remain unaffected by continuous humidity and by chemicals present in a vehicle exhaust type of air duct environment. Anchor rods shall be recommended by manufacturer for highest vibration resistant application. Anchors shall be heat tested as follows:

- 1. A minimum of 3 anchors of each anchor size used on the project shall be heat tested. Anchors shall be installed in a concrete block measuring 1 foot x 1 foot x 1 foot to a depth matching design embedment depth. Concrete for the test block shall have a 28 day compressive strength of 4000 psi. Anchor and surface on which anchor is installed shall be subject to the specified temperature rise. The remaining 5 sides of the concrete block need not be exposed to the temperature rise. Anchor shall be tension loaded to a service design service load of 3.25 kips tension. The temperature of the anchor environment shall then be increased to a level of 1,500 degrees F within a period no longer than 10 minutes. The temperature shall be maintained at 1,500 degrees F for an additional 30 minute period. Should any failure of the bond between the anchor and the concrete occur, including any differential movement between anchor and concrete, during any of the tests, such failure shall be sufficient cause for rejection of the anchor. All tests shall be conducted in the presence of the Engineer.”*

The contractor transmittals from Modern Continental Company (MCC) to Bechtel / Parsons Brinckerhoff for the Structural Calculations for Adhesive Anchors was submitted under the correct Specification Number 723.480.

MCC entered into a purchase agreement with Newman, Renner, and Colony (NRC) that was signed by MCC on June 3, 1999. The purchase agreement included a description of work and prices for furnishing fasteners, hangers, bolts, and miscellaneous metals. The purchase agreement included a description, quantity, unit price, and total price for the fasteners, hangers, bolts, and miscellaneous metals. The purchase agreement did not specify the manufacturer of the anchor.

NTSB investigators obtained a fax dated March 16, 1999 that appeared to be generated by MCC that included a list of fasteners, hangers, and bolts with the unit prices left blank. The unit prices were filled in by handwritten notation by NRC. The fax did not specify the manufacturer of the anchor. Powers Fasteners (formerly known as Rawlplug Co.), from at least 1991 through 1995, had an office at the same location as NRC located at 190 University Avenue, Westwood, MA 02090. This is the same location stated as NRC's office address in the purchase agreement signed on June 3, 1999.

NRC hired the firm of Simpson, Gumpertz, and Heger, Inc. to obtain approval from the Massachusetts Highway Department for use of the NRC-1000 Gold Epoxy.

Powers Fasteners sent a letter to Simpson, Gumpertz, and Heger, Inc. dated June 3, 1999 indicating *"NRC-1000 Gold is the identical product to the Power-Fast Epoxy which we currently sell. Catalog No. 8431 is the fast set material while Catalog No. 8433 is the standard setting material."*

Powers Fasteners sent a letter to Simpson, Gumpertz, and Heger, Inc. dated July 23, 1999 transmitting a creep test report for the Power-Fast Slow Set Adhesive *"Power-Fast Slow Set Adhesive – Creep Test (Report No. 6R25).....We hope this information assists you in your pursuit of State of Massachusetts Highway Approval of Power-Fast / NRC-1000 Gold Premier Epoxy."*

Powers Fasteners sent a letter to Simpson, Gumpertz, and Heger, Inc. dated November 15, 1999 transmitting a binder that contained a complete collection of independent test data regarding the Power-Fast Epoxy Injection Gel / NRC-1000 Gold Premier Epoxy. The last paragraph in the letter indicated *"Hopefully, this information will be helpful in your pursuit of State of Massachusetts Highway Department Approvals."*

Simpson, Gumpertz, and Heger, Inc. sent a letter to the Massachusetts Highway Department dated November 16, 1999 that indicated their client, NRC, wished to obtain approval for the NRC-1000 Gold Epoxy. The letter transmitted a draft copy of an ICBO Report that was intended to assist in the evaluation of the epoxy.

"As we discussed, our client, Newman Renner Colony, wishes to obtain approval of the Massachusetts Highway Department. To assist you in your evaluation, we have enclosed a copy of the draft copy ICBO Report for the Power-Fast Adhesive Anchor System and a collection of the independent test data. Note that NRC-1000 GOLD is the same product as the Power-Fast Epoxy, as documented on the enclosed letter. Also enclosed is the NRC-1000 GOLD Design Manual."

Simpson, Gumpertz, and Heger, Inc. sent a transmittal to the Massachusetts Highway Department dated March 20, 2000 that enclosed a copy of the ICBO Report for the Power-Fast Adhesive Anchor System. The cover letter attached to the ICBO Report indicated *“The enclosed evaluation report draft, dated February 1, 2000, has received final approval and will be released shortly for publication and distribution.”*

Powers Fasteners sent an internal memorandum to NRC dated April 3, 2000 indicating that all testing had been completed by the Massachusetts Highway Department and the NRC-1000 Gold Epoxy System had been approved. The internal memorandum indicated that Powers Fasteners was waiting for a letter from the State formally announcing its approval and suggested NRC apply the appropriate pressure on the State. The internal memorandum also recommended that NRC actively market the product and its approval and this may serve as a catalyst needed for obtaining a letter from the State.

“The purpose of this memo is to bring you up to date as to the approval status of the NRC 1000 Gold Epoxy with the Mass Highway Department.

To date, all information has been submitted by SGH and all testing has been completed by the State of Mass. The NRC 1000 Gold Epoxy System has been approved.

However, technically we are waiting for a letter from the State formally announcing its approval. With regard to this I would suggest that (employee of NRC) keep in close contact with (employee of SGH) of SGH to apply the appropriate pressure on the state.

In the meantime, I would recommend that NRC actively market the product and its approval. This may also serve as the catalyst needed for this letter.”

The NRC-1000 Gold Epoxy and Power-Fast Epoxy appeared on the state’s qualified list dated May 19, 2000. The NRC-1000 Gold Epoxy was used in the I-90 eastbound connector tunnel from June 1999 through March 2000 without being on the Massachusetts Highway Department state’s qualified list.

Dan Walsh /s/

Dan Walsh
Highway Accident Investigator